

MongoDB Database Schema

Collection definitions, field types, constraints, and example documents

Database: hr_management_db Collections: 7 Engine: MongoDB 6+ · Spring Data MongoDB

Overview

Collection	Purpose	Auto-seeded
users	Login credentials — email, hashed password, role	Admin on first startup
employees	Full employee profiles linked to users and departments	No
departments	Department records with manager and HR assignment	DEPT-0000 on first startup
leave_requests	Individual leave applications with approval status	No
leave_balances	Remaining leave quota per employee (4 leave types)	Auto-created on registration
payroll	Monthly payroll records with net salary	No
employee_history	Immutable audit trail — promotions, designation changes, notes	No

Collection Schemas

users

One document per registered user. Passwords are BCrypt-hashed and never returned by any API endpoint.

```
{
  "_id": "ObjectId",
  "email": "admin@hrms.com",
  "password": "$2a$10$...", // BCrypt hash – never exposed
  "role": "ADMIN",
  "createdAt": "2026-09-03T01:50:00",
  "updatedAt": "2026-09-03T01:50:00"
}
```

Field	Type	Required	Notes
_id	ObjectId	Auto	MongoDB default primary key
email	String	Yes	Unique. Lookup is case-insensitive at login.
password	String	Yes	BCrypt-hashed. Never returned by API.
role	Enum	Yes	ADMIN EMPLOYEE HR
createdAt	DateTime	Auto	Set on registration
updatedAt	DateTime	Auto	Updated on every change

employees

Full profile for each employee. Linked to users by email and to departments by departmentId .

```
{
  "_id": "ObjectId",
  "employeeId": "EMP-0001",
  "name": "John Doe",
  "email": "john.doe@company.com",
  "phone": "+92-300-1234567",
  "gender": "MALE",
  "dateOfBirth": "1995-06-15",
  "address": "123 Main Street, Karachi",
  "departmentId": "DEPT-0001",
  "designation": "Software Engineer",
  "dateOfJoining": "2026-01-01",
  "salary": 80000,
  "allowances": 10000,
  "deductions": 5000,
  "employeeType": "EMPLOYEE",
  "userRole": "EMPLOYEE",
  "status": "ACTIVE",
  "createdAt": "2026-09-03T10:00:00",
  "updatedAt": "2026-09-03T10:00:00"
}
```

Field	Type	Required	Notes
employeeId	String	Auto	Sequential: EMP-0001, EMP-0002, ...
name	String	Yes	Full name
email	String	Yes	Must match <code>users.email</code> for the same person
phone	String	Yes	
gender	Enum	Yes	MALE FEMALE
dateOfBirth	Date	Yes	
address	String	Yes	
departmentId	String	Yes	References <code>departments.departmentId</code> . HR employees default to DEPT-0000 until assigned.
designation	String	Yes	Job title. Changing it auto-creates a DESIGNATION_CHANGE history record.
dateOfJoining	Date	Yes	
salary	Number	Yes	Base monthly salary
allowances	Number	No	Added to net salary in payroll
deductions	Number	No	Subtracted from net salary in payroll
employeeType	Enum	Yes	EMPLOYEE MANAGER HR
userRole	Enum	Yes	EMPLOYEE HR — determines which dashboard the person sees
status	Enum	Auto	ACTIVE INACTIVE . Defaults to ACTIVE.

departments

One document per department. Every department must have a dedicated HR employee assigned.

```
{
  "_id":          "ObjectId",
  "departmentId": "DEPT-0001",
  "name":         "Engineering",
  "description":  "Software development team",
  "managerId":   "EMP-0002",  // optional
  "hrId":        "EMP-0003",  // required — one HR per dept
  "isSystem":    false,
  "createdAt":   "2026-09-03T11:00:00",
```

```
"updatedAt": "2026-09-03T11:00:00"
}
```

Field	Type	Required	Notes
departmentId	String	Auto	Sequential: DEPT-0001, DEPT-0002, ...
name	String	Yes	Unique department name
description	String	No	
managerId	String	No	References employees.employeeId . Must be MANAGER type.
hrId	String	Yes	References employees.employeeId . Must be HR role. One HR per department only — uniqueness enforced at service layer.
isSystem	Boolean	Auto	true only for DEPT-0000 (Unassigned). System departments cannot be deleted.

Auto-seeded system department: DEPT-0000 "Unassigned" — a holding bucket for employees without a department. Created automatically on first startup.

leave_requests

One document per leave application submitted by an employee.

```
{
  "_id": "ObjectId",
  "leaveId": "LVE-0001",
  "employeeId": "EMP-0001",
  "leaveType": "ANNUAL",
  "startDate": "2026-10-01",
  "endDate": "2026-10-05",
  "totalDays": 5,
  "reason": "Family vacation",
  "status": "PENDING",
  "appliedAt": "2026-09-15T09:00:00",
  "updatedAt": "2026-09-15T09:00:00"
}
```

Field	Type	Notes
leaveId	String	Sequential: LVE-0001, LVE-0002, ...
employeeId	String	References employees.employeeId

Field	Type	Notes
leaveType	Enum	ANNUAL CASUAL SICK UNPAID
startDate	Date	Must not be after endDate
endDate	Date	
totalDays	Number	Auto-calculated: endDate - startDate + 1
reason	String	
status	Enum	PENDING → APPROVED or REJECTED . Only PENDING requests can be actioned by HR.

leave_balances

One document per employee. Auto-created with default values when an employee profile is registered.

```
{
  "_id": "ObjectId",
  "employeeId": "EMP-0001",
  "annualLeaves": 18,
  "casualLeaves": 10,
  "sickLeaves": 8,
  "unpaidLeaves": 5,
  "updatedAt": "2026-09-03T10:00:00"
}
```

Each quota field is automatically decremented when a leave request of that type is **approved**. If the balance is insufficient, the approval is rejected with a validation error.

payroll

One document per employee per month/year. Duplicate payroll for the same employee + month + year is not allowed.

```
{
  "_id": "ObjectId",
  "payrollId": "PAY-0001",
  "employeeId": "EMP-0001",
  "month": 9,
  "year": 2026,
  "basicSalary": 80000,
  "allowances": 10000,
  "deductions": 5000,
  "netSalary": 85000, // basicSalary + allowances - deductions
  "paymentStatus": "UNPAID",
}
```

```
"generatedAt": "2026-09-30T18:00:00",
"updatedAt": "2026-09-30T18:00:00"
}
```

Field	Notes
basicSalary	Copied from <code>employees.salary</code> at generation time
allowances	Copied from <code>employees.allowances</code>
deductions	Copied from <code>employees.deductions</code>
netSalary	Auto-calculated: <code>basicSalary + allowances - deductions</code>
paymentStatus	Starts as <code>UNPAID</code> . HR marks it <code>PAID</code> .

employee_history

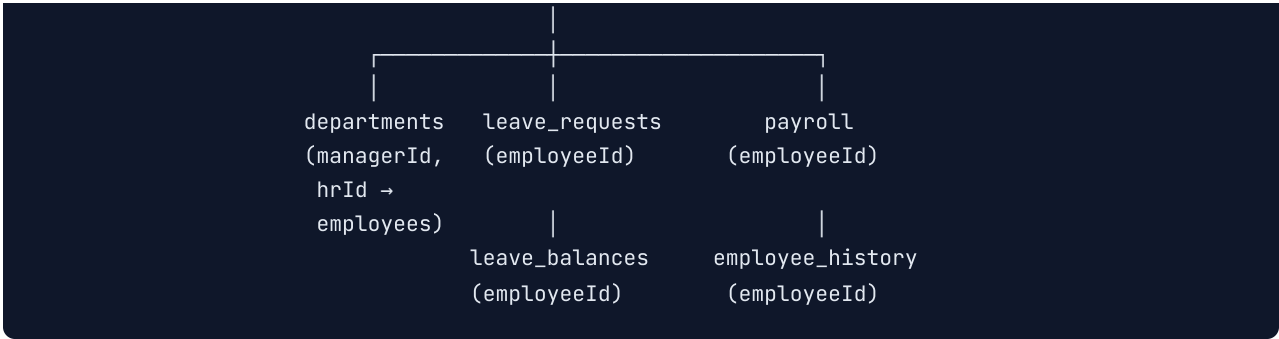
Immutable audit log. Records are never deleted — they capture every significant event in an employee's career.

```
{
  "_id": "ObjectId",
  "employeeId": "EMP-0001",
  "type": "PROMOTION",
  "previousValue": "EMPLOYEE",
  "newValue": "MANAGER",
  "note": "Promoted to Manager role",
  "recordedAt": "2026-09-10T14:30:00"
}
```

Trigger	Type	Previous / New value
Admin changes employee type	PROMOTION	Old type → New type (e.g. EMPLOYEE → MANAGER)
Admin changes designation via profile update	DESIGNATION_CHANGE	Old designation → New designation
Admin adds a performance note	PERFORMANCE_NOTE	Note text stored in <code>note</code> field

Entity Relationships

```
users ————— employees
(email links the two)      (employeeId)
```



MongoDB does not enforce foreign key constraints. Referential integrity is maintained at the application (service) layer. For example, the service validates that a department's `hrId` references an existing HR-type employee before saving.

Recommended Indexes (Production)

Collection	Field(s)	Type	Reason
users	email	Unique	Login lookup
employees	employeeId	Unique	Profile lookup by ID
employees	email	Unique	Prevents duplicate profiles
employees	departmentId	Regular	Department scoping queries
departments	departmentId	Unique	Department lookup
departments	hrId	Unique	Enforce one HR per department
leave_requests	employeeId	Regular	Fetch employee's leave history
leave_requests	status	Regular	Filter by status
leave_balances	employeeId	Unique	One balance per employee
payroll	employeeId	Regular	Payroll history per employee
payroll	(employeeId, month, year)	Compound Unique	Prevent duplicate monthly payroll
employee_history	employeeId	Regular	History timeline per employee

